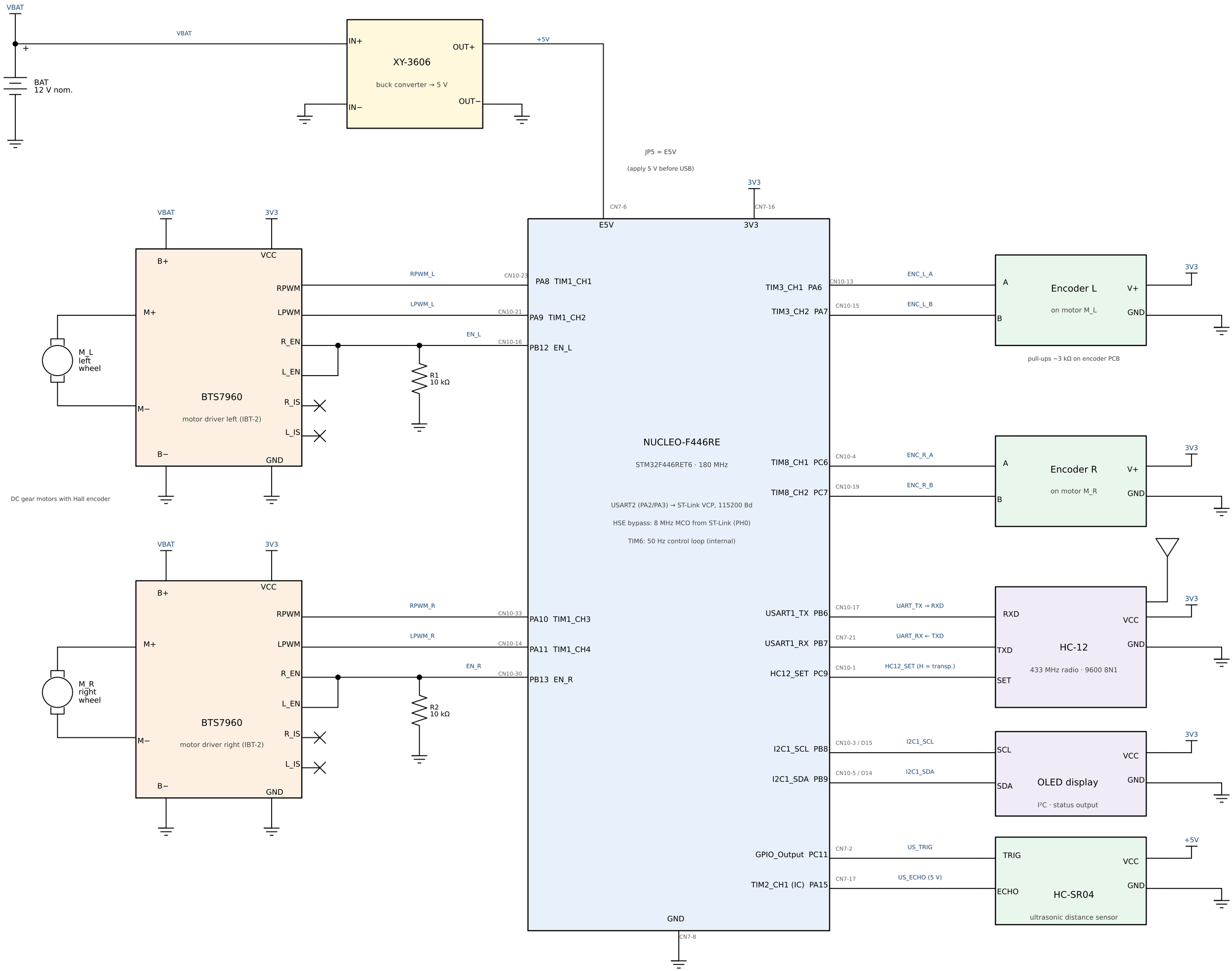

Notes

1 All grounds are common: battery −, XY-3606, BTS7960, Nucleo, encoders, HC-12, OLED, HC-SR04.

2 Nucleo is powered from XY-3606 5 V into E5V (CN7-6), JP5 on E5V. Do not use VIN.

3 Nucleo 3V3 (CN7-16) supplies HC-12, OLED, BTS7960 logic VCC, encoder V+.
Only one 3.3 V source (Nucleo 3V3 or AMS1117, never both).

4 R_EN and L_EN of each BTS7960 are bridged. 10 kΩ pull-downs (plus internal pull-down) keep the drivers disabled during reset and programming.

5 TIM1 PWM 20 kHz. Encoders in TIM3/TIM8 encoder mode (x4), input filter 15.

6 USART1 TX/RX are crossed to HC-12 RXD/TXD. PB7 internal pull-up (to verify).

7 R_IS / L_IS current-sense outputs are not connected.

8 OLED on I2C1 (PB8 SCL / PB9 SDA = Arduino D15 / D14). I²C pull-ups are usually on the OLED module; if not, add 4.7 kΩ from SCL and SDA to 3V3.

9 HC-SR04 runs on +5V (XY-3606). Echo is a 5 V signal into PA15 (5 V tolerant), TIM2_CH1 input capture on the 1 MHz time base. Trig from PC11 (3.3 V logic).

MR2006B Industrial Automation · Activity 2.6

Car wiring schematic

Board: NUCLEO-F446RE · Project: Challenge_TestHC

Source: CubeMX configuration (status 09.10.2026)

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